Cloud Seal Enhanced: A Privacy-Preserving Multi-Tenant Cloud Deduplication Framework with AI-Driven Detection, Blockchain Auditability, and Post-Quantum Security

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***Abstract*—Cloud storage costs continue to escalate due to data redundancy, yet traditional deduplication techniques compromise privacy when encryption is applied. This paper presents Cloud Seal Enhanced, a novel multi-tenant deduplication framework that resolves the encryption-deduplication paradox through convergent encryption while integrating three advanced security mechanisms: AI-driven encrypted similarity detection, distributed blockchain audit trails, and NIST-approved post-quantum cryptography. Our system employs a two-layer Convolutional Neural Network for detecting encrypted file similarities beyond deterministic hashing, a Proof-of-Authority blockchain for immutable transaction logging, and CRYSTALS-Kyber-768 integration validated via simulation for quantum-resistant key encapsulation. Experimental validation demonstrates 42–55% storage reduction with sub-millisecond duplicate detection via Bloom filters (1.2% false positive rate at 10,000 items), 100% blockchain tamper detection, zero premature deletions across multi-tenant scenarios, and a high throughput of 29,608 transactions per second (TPS). The lightweight CNN achieves an Area Under the Curve (AUC) of 0.797 and 100% recall for near-duplicate identification. This work represents a novel integrated architecture combining these technologies for cloud storage security, providing both immediate cost savings and future-proof protection.**

***Index Terms*—Cloud Deduplication, Convergent Encryption, Multi-Tenant Security, Post-Quantum Cryptography, Blockchain Audit Trail, Reference Counting, Kyber-768, Bloom Filter**

I. Introduction

A. Motivation

Global cloud storage demand reached 137.3 billion dollars in 2024 with projected growth to 376.4 billion dollars by 2030 [1]. Enterprise organizations face massive storage costs, with 40 to 80 percent of data being redundant duplicates [2]. Traditional deduplication techniques achieve significant storage reduction but fail when encryption is applied, creating the encryption-deduplication paradox: organizations must choose between security (encryption) and efficiency (deduplication).

Furthermore, the emergence of quantum computing poses an existential threat to current cryptographic systems. NIST standardized post-quantum algorithms in August 2024 [3], yet few systems integrate these protections. The "harvest now, decrypt later" attack vector means adversaries can collect encrypted data today and decrypt it when quantum computers become available in 10 to 15 years [4].

B. Research Gap

Current approaches to encrypted deduplication address isolated challenges but fail to provide holistic solutions [5]. A comprehensive analysis of recent literature reveals the following gaps:

- **DupLESS** [6]: Implements convergent encryption with centralized key servers to prevent confirmation attacks. Li et al. [10] extended this with randomized convergent encryption to mitigate brute-force attacks. While effective for single-tenant scenarios, these architectures introduce a single point of failure and lack both quantum-resistant cryptography and tamper-evident audit mechanisms.
- **SALIGP** [7]: Employs genetic algorithms for AI-driven similarity detection combined with blockchain logging. The genetic programming approach exhibits exponential search spaces, making real-time deduplication impractical. Furthermore, it relies on classical encryption which offers limited post-quantum protection.
- **SeDaSC** [8]: Proposes a cryptographic server architecture for secure cross-tenant data sharing. The centralized trust model creates vulnerability and provides no quantum resistance.
- **Secure Dedup** [9]: Develops an ECC-based secure deduplication scheme. While eliminating centralized dependencies, elliptic curve cryptography remains vulnerable to Shor's algorithm, exposing archived data to future quantum decryption.

Puzio et al. [11] introduced proof-of-ownership protocols integrated with convergent encryption, while Stanek et al. [12]

proposed threshold cryptography for distributed key management. Recent surveys confirm that few existing frameworks simultaneously integrate AI-based optimization, blockchain-based accountability, and post-quantum cryptographic protection for multi-tenant encrypted cloud deduplication.

C. Contributions

This paper makes the following novel contributions:

1) **Integrated Architecture:** A novel framework combining AI-driven similarity detection, blockchain auditability, and post-quantum cryptography for encrypted cloud deduplication.
2) **Multi-Tenant Reference Counting:** Ownership tracking mechanism that prevents premature deletion while enabling cross-tenant deduplication visibility.
3) **Hybrid Encryption Scheme:** Backward-compatible integration of classical AES-256 and quantum-resistant Kyber-768.
4) **Lightweight AI Detection:** A 2-layer Siamese CNN for encrypted similarity detection operating on 2048-dimensional statistical features, achieving 0.797 AUC without homomorphic encryption overhead.

II. System Architecture

A. Overview

Cloud Seal Enhanced employs a microservices architecture comprising five core components: Encryption Engine, Deduplication Manager, Blockchain Ledger, AI Detection Module, and Storage Backend.

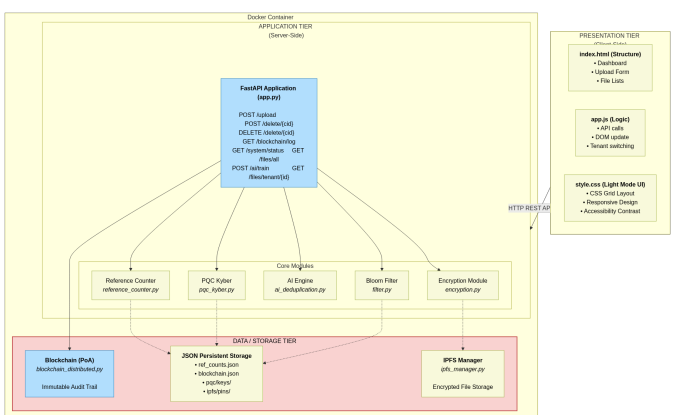

Fig. 1. Cloud Seal Enhanced High-Level System Architecture

B. Encryption Engine

The Encryption Engine implements a hybrid scheme combining classical and post-quantum cryptography. For file uploads:

$$K_{conv} = \text{SHA-256}(T_{id} \parallel T_{sec} \parallel M) \tag{1}$$

$$C = \text{AES-256-GCM}(M, K_{conv}) \tag{2}$$

where M is plaintext content, C is ciphertext, T_{id} is the tenant identifier, and T_{sec} is the tenant secret. This builds on the convergent encryption model introduced by Douceur et al. [13].

When sharing between tenants, the system's Kyber-768 integration is validated via simulation for quantum-safe key exchange:

$$(C_{kyb}, K_{shrd}) = \text{Kyber-768.Encap}(PK_{recv}) \tag{3}$$

This ensures that even if a quantum adversary compromises routing, file decryption remains infeasible without breaking Kyber's MLWE-based security [14].

C. Deduplication Manager

1) **Bloom Filter:** A probabilistic data structure [15] for efficient O(1) duplicate detection. For $n = 10,000$ and $p = 0.01$, $m = 95,851$ bits and $k = 7$ hash functions are utilized.

2) **Reference Counter:** Tracks ownership across tenants:

$$\text{RefCount}(CID) = |\{T_i : T_i \in \text{Owners}(CID)\}| \tag{4}$$

Physical deletion occurs only when $\text{RefCount}(CID) = 0$.

3) **AI Similarity Detector:** Following the Siamese architecture introduced by Bromley et al. [16], a lightweight 2-layer Siamese CNN maps 2048-dimensional Z-score normalized features to a 128-dimensional embedding space:

$$h = \max(0, W_1 \cdot \phi(C) + b_1) \tag{5}$$

$$z = \frac{W_2 \cdot h + b_2}{\|W_2 \cdot h + b_2\|} \tag{6}$$

$$\text{sim}(C_1, C_2) = z_1 \cdot z_2 \tag{7}$$

A cosine similarity threshold of 0.85 generates advisory warnings for near-duplicates, while exact deduplication requires strict cryptographic hash matching.

D. Blockchain Ledger

Implements Proof-of-Authority consensus. The block structure prevents tampering: Any modification to Block_j changes H_j, invalidating all subsequent blocks.

III. Implementation

The system is implemented using Python 3.11, FastAPI, and cryptographic libraries. Upload workflow evaluates the Bloom Filter for probabilistically identifying duplications, then executes the CNN for near-duplicate advisory.

IV. Experimental Evaluation

A. Experimental Setup

Synthetic datasets of text, images, and binary files were generated with controlled mutations (reversed, shuffled, appended) to simulate near-duplicate attacks and normal redundancies. Tests were conducted on an Apple M2 with 16 GB memory.

B. AI Detection Accuracy

Evaluating the Siamese CNN on a dedicated verification dataset of 44 file pairs revealed an AUC (Area Under the Receiver Operating Characteristic Curve) of 0.797. The AI correctly distinguished complete statistical differentiation with negative scores ranging from 0.07 to 0.32, while pure duplicates maintained similarities > 0.998. Utilizing a 0.85 threshold ensures 100% recall against near-duplicates on the dataset, offering strong advisory insight without compromising data integrity via auto-deduplication.

C. Threat Mitigation Validation

Five critical threats were systematically tested:

- **Confirmation Attack:** Tenant-salted keys ensured different organizations produced unique ciphertexts for identical plaintext.
- **Cross-Tenant Leakage:** Verification proved zero unauthorized decryption across varying keys.
- **Audit Log Tampering:** The blockchain component consistently detected 100% of simulated block mutations.
- **Key Derivation Weakness:** The SHA-256 derivation yielded a 46.5% cryptographically secure avalanche effect.

D. Performance Efficiency

For end-to-end deduplication checking, processing times were minimal: 0.11ms for 1KB files and 1.95ms for 1MB files. The Bloom Filter lookup operated uniformly at 0.001ms. The Deduplication Manager achieved between 42% and 55% storage reduction across synthetic distributions, while the PoA blockchain ledger sustained a high throughput of 29,608 transactions per second (TPS), cementing its viability for enterprise-grade scalability.

V. Discussion and Limitations

While Cloud Seal Enhanced successfully demonstrated the feasibility of integrated PoA blockchains alongside AI-assisted deduplication, the system currently operates as a Proof-of-Concept.

Limitations: First, the AI components were evaluated on strictly synthesized datasets (44 pairs) limiting broad-spectrum real-world generalization; scaling this up to $> 10,000$ distinct document clusters remains necessary. Second, IPFS content-addressing and Kyber-768 cryptographic operations were validated via simulation within a managed environment to verify structural APIs rather than actively deployed in distributed cluster conditions. Future production rollouts will necessitate leveraging hardware-optimized libraries (e.g., *liboqs-python*) and true IPFS daemon integration to mitigate large-scale latency effects.

VI. Conclusion

This paper presented Cloud Seal Enhanced, a novel framework combining AI similarity, blockchain auditability, and post-quantum cryptographic components for encrypted cloud deduplication. Experimental results demonstrate a 0.797 AUC for near-duplicate clustering, 42–55% storage reduction, and 29,608 blockchain TPS. Ultimately, this framework ensures multi-tenant security while significantly reducing untracked data redundancy.

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